The Impact of BTI Variations on Timing in Digital Logic Circuits

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***Abstract*—A new framework for analyzing the impact of bias temperature instability (BTI) variations on timing in large-scale digital logic circuits is proposed in this paper. This approach incorporates both the reaction–diffusion model and the charge-trapping model for BTI and embeds these into a temporal statistical static timing analysis framework capturing process variations and path correlations. Experimental results on 32-, 22-, and 16-nm technology models, which were verified through Monte Carlo simulation, confirm that the proposed approach is fast, accurate, and scalable and indicate that BTI variations make a significant contribution to circuit-level timing variations.**

***Index Terms*—Bias temperature instability (BTI), circuit reliability, process variation (PV), timing analysis.**

I. Introduction

RELIABILITY issues in very large scale integrated (VLSI) circuits have been a growing concern as technology trends in semiconductor technologies show progressive downscaling of feature sizes. One of the major reliability issues is bias temperature instability (BTI), which causes the threshold voltage $V_{\rm th}$ of CMOS transistors to increase over time under voltage stress, resulting in a temporally dependent degradation of digital logic circuit delay. Various optimizations have been proposed to cope with this degradation, such as slowing the operating frequency with time, adding delay guardbands, and using adaptive methods to recover from delay degradation.

The reaction–diffusion (R–D) model [1]–[3], based on dissociation of Si–H bonds at the Si/SiO_2 interface, has been the prevailing theory of BTI mechanism and has been widely used in research on circuit optimization and design automation. However, over the years, several limitations in the theory have been exposed. For instance, in R–D theory, the rate of recovery is determined by the diffusion of neutral hydrogen atoms, which is not affected by the gate bias. However, the measured device recovery begins faster and lasts longer than the prediction of R–D theory and shows strong dependence on the applied gate bias. An alternative mechanism for explaining BTI effects is the charge-trapping (CT) and detrapping model [4], in which the defects in gate dielectrics can capture charged carriers, resulting in $V_{\rm th}$ degradations. The major difference between the two models is the nature of the diffusing species and the medium that facilitates the diffusion. Based on published works, both R–D and CT mechanisms exist in current semiconductor technologies, and the superposition of both models is shown to better match experimental device data [3].

In nanometer-scale technologies, variations in the BTI effect are gaining a great deal of attention under both R–D and CT frameworks due to the random nature of defect localization in smaller and smaller transistors; together, these result in increased variations in the number of defects in a transistor. While there has been a great deal of research on timing variability due to process variations (PVs) [5]–[7], and a few previous works have combined random variation effects from PVs with deterministic BTI degradations [8]–[10], the problem of BTI variations has not received much attention.

Most of the published circuit-level works incorporating BTI variations are based on the variability model of $\Delta N_{\rm IT}$ randomness within the R–D framework, introduced in [11]. This model was applied to analytically determine the effect of BTI variations on SRAM and logic cells, and on circuit and pipeline performance using Monte Carlo (MC) simulations in [12] and [13]. However, as explored in our paper, for digital logic circuits, the $\Delta N_{\rm IT}$ variation in this R–D based model has a relatively small impact on circuit timing variation, as compared with variations under the CT model and PVs. Another model of the BTI-related variations was considered in [9], as caused by process perturbations. Since these small model perturbations lead to a relatively small change in the BTI-driven delay shift, the impact on circuit timing is a second-order perturbation that is relatively small.

On the other hand, the variations of device-level BTI degradations under the CT model has been discovered to be a significant issue for nanoscale transistors. CT and detrapping at each defect are random events that are characterized by the capture and emission time constants. This paradigm is intrinsically statistical, and it captures not only the variations in the number of defects, but also the variations in $\Delta V_{\rm th}$ induced by each defect [14]–[16]. Under this statistical model, the variation of device lifetime significantly increases, particularly for devices with a smaller number of defects N, as illustrated in Fig. 1.

However, the impact of BTI variations under the CT model on circuit performance has not received much attention, with only limited works that explore this issue beyond the device level. In [17], models and approaches were proposed for analyzing the impact of BTI variations on circuit performance; however, the proposed SPICE-based atomistic approaches are time consuming and not scalable to large-scale circuits. As we will show, the CT model is the dominant component of

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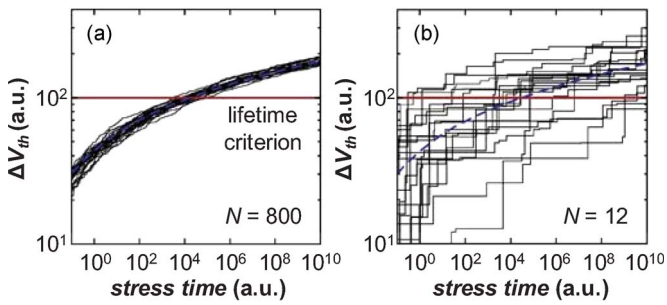

Fig. 1. (a) Narrow distribution of lifetime in large devices where randomness averages out. (b) Large variation of lifetime in small devices where stochasticity predominates [17].

BTI intrinsic variations and makes significant contributions to circuit delay variation. Furthermore, its impact rapidly grows as devices scale down, posing increasingly severe reliability issues to digital logic circuits.

In this paper, we first introduce the notion of precharacterized *mean defect occupancy probability* for the CT model to effectively reduce the complexity of circuit-level analysis and to make it possible to handle large-scale circuits. Then, we incorporate variations under both the R–D model and the CT model into a novel temporal statistical static timing analysis (T-SSTA) framework, capturing randomness from both PVs and temporal BTI degradations. We exercise this approach on large digital logic circuits and show simulation results for the 32-, 22-, and 16-nm technology nodes. The correlation of process parameters due to path reconvergence is efficiently considered in modeling and analysis to guarantee both high accuracy and low complexity. To the best of our knowledge, this is the first circuit-level work that incorporates variations in BTI effects into SSTA under a scalable and computationally efficient procedure.

Our experimental results are based on simulations and show that the proposed analysis approach has an accuracy that lies within 2.2% of MC simulation while speeding up the calculation by 15×. Averaging over all benchmarks considered in this paper, the fraction of the variance attributable to PVs, BTI R–D effects, and BTI CT effects are 81%, 3%, and 16% at the 32-nm node; 70%, 4%, and 26% at the 22-nm node; and 66%, 5%, and 29% at the 16-nm node, respectively. Thus, under these models, the relative role of BTI CT to circuit variability is projected to significantly increase in the future but is less than the contribution of PVs.

II. Modeling Variations

This section introduces the models used to capture the effects of variations that affect BTI-induced aging. We begin by discussing BTI variations under both the R–D and CT models. Next, we overview models for PVs, including spatial correlation effects. As in [3], the total threshold degradation $\Delta V_{\rm th}$ of an MOS device is modeled by superposition as

$$\Delta V_{\rm th} = \Delta V_{\text{th-RD}} + \Delta V_{\text{th-CT}} + \Delta V_{\text{th-RDF}} \tag{1}$$

in which the BTI terms $\Delta V_{\text{th-RD}}$ and $\Delta V_{\text{th-CT}}$ are independent Gaussian random variables that will be given in (5) and (15), and $\Delta V_{\text{th-RDF}}$ is the variation component due to random dopant fluctuation (RDF) [18], [19], which also follows Gaussian and have no spatial correlations [20]. For each transistor, the sum, i.e., $\Delta V_{\rm th}$, of Gaussian variables is still a Gaussian, and this sum is an independent random variable for different MOS transistors.

A. BTI Variability Under the R–D Model

Under the R–D framework, the mechanism of BTI in a MOS transistor is explained through the dissociation of Si–H bonds at the Si/SiO_2 interface and the diffusion of hydrogen into dielectric and gate. The number of generated interface traps, i.e., Si^+, is denoted as $\Delta N_{\rm IT}$, and absolute value of the induced threshold voltage shift, i.e., $\Delta V_{\rm th}$, is

$$\Delta V_{\rm th} = \frac{q\Delta N_{\rm IT}}{C_{\rm ox}}. \tag{2}$$

Under the R–D model, the long-term threshold voltage shift $\Delta V_{\rm th}$ under ac BTI stress is modeled in [12] and [21] as

$$\Delta V_{\rm th}^{\rm (nom)} = f_{\rm AC}(SP) \cdot K_{\rm DC} \cdot t^n \tag{3}$$

in which $K_{\rm DC}$ is a technology-dependent constant for dc BTI degradation, and $f_{\rm AC}(SP)$ is the coefficient that captures the ac degradation with signal probability SP (the probability of effective BTI stress). The function $f_{\rm AC}(SP)$ can be numerically precomputed using method proposed in [2].

For deeply scaled technologies, the device size is small enough that $\Delta N_{\rm IT}$ is a random variable, modeled as a Poisson distribution [11], i.e.,

$$\Delta N_{\rm IT} \sim \text{Poisson}(\lambda),$$
$$\text{where } \lambda = \Delta N_{\rm IT}^{\rm (nom)} = \Delta V_{\rm th}^{\rm (nom)} \cdot C_{\rm ox}/q. \tag{4}$$

Our reliability analysis focuses on late lifetime behavior, when the average numbers of interface traps λ in MOS transistors have relatively large values. For instance, the value of λ corresponding to $\Delta V_{\rm th} = 0.1$ V for a device with $(W/L) = 2$ is about 49 for 32-nm PTM [22] model or 15 for 16-nm PTM model, and it proportionally increases with the device size. It is well known that, for $\lambda > 10$, a Gaussian approximates the Poisson distribution well [23]. Therefore, to simplify our analysis without significant loss of accuracy, this Poisson distribution is approximated as a Gaussian distribution with the same mean and variance $\mu = \sigma^2 = \lambda$, hence $\Delta N_{\rm IT} \sim N(\lambda, \lambda)$. In (2), the threshold voltage degradation under R–D model has the distribution

$$\Delta V_{\text{th-RD}} \sim N\left(\frac{q\lambda}{C_{\rm ox}}, \frac{q^2\lambda}{C_{\rm ox}^2}\right). \tag{5}$$

As will be shown in Section IV, this distribution approximation does not induce significant errors to the circuit-level results.

B. BTI Variability Under the CT Model

Recent work [4] on the BTI effect of small-area devices has revealed that the degradation and recovery of $\Delta V_{\rm th}$ proceed

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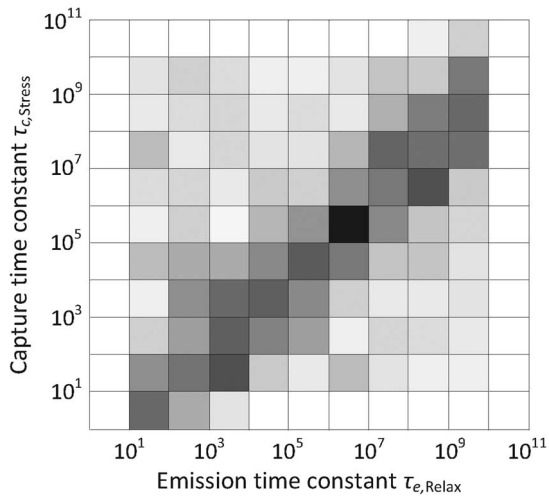

Fig. 2. Distribution of defects according to their capture time constant τ_c and emission time constant τ_e.

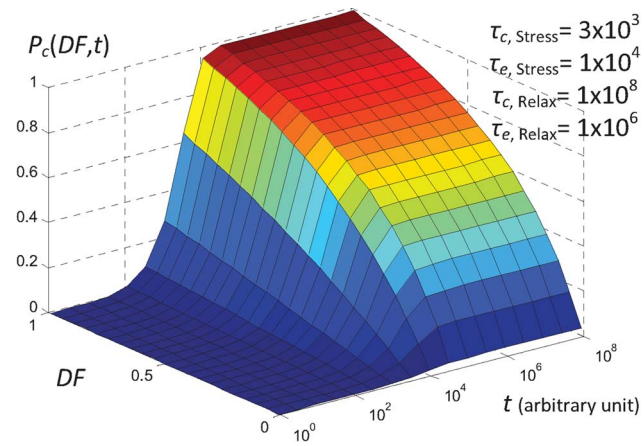

Fig. 3. Example plot of defect occupancy probability function $P_c(DF, t, \vec{\tau})$ of a single defect.

in discrete steps, with variable heights, which could not be explained by the R–D model but are fully consistent with CT, which is also observed in random telegraph noise and $1/f^2$ noise.

Based on these observations, a newer CT model was proposed for the BTI effect, in which each defect is characterized by parameters of the capture time τ_c and emission time τ_e and each defect's contribution to the device threshold change ΔV_t. These parameters are characterized using the time-dependent defect spectroscopy method [4], [14], as a distribution shown in the form of a density map as Fig. 2, in which defects with similar time constants are binned together, and the total ΔV_t is shown in each grid.

If this characterization is performed on a large enough device, with the assumption that ΔV_t of all defects are independent and identically distributed (i.i.d.), the density map could be interpreted as the distribution of defects, in which each grid's value represents the probability of defects falling into that grid. The generation of this distribution is part of technology process characterization and independent of circuit structure.

CT (capture) and detrapping (emission) is a stochastic process. Following the models in [16], the capture time τ_c and the emission time τ_e are strongly dependent on bias voltage and temperature. In digital circuits, there are only two nontransient voltage stages, i.e., logic 1 and logic 0; hence, the bias condition can be simplified to two static modes of stress and relaxation. We capture the temperature dependence effect by the use of a standard corner-based approach where the worst case temperature corner is assumed. This way, each defect can be described by four time constants, denoted by the vector $\vec{\tau}$ as

$$\vec{\tau} = (\tau_{c,\mathrm{Stress}}, \tau_{c,\mathrm{Relax}}, \tau_{e,\mathrm{Stress}}, \tau_{e,\mathrm{Relax}}). \tag{6}$$

The defect occupancy probability (i.e., the probability of CT) of a single defect with time constants $\vec{\tau}$ under ac stress of duty factor DF and time span t is derived in [16] to be

$$P_c(DF, t, \vec{\tau}) = \frac{\tau_e^*}{\tau_c^* + \tau_e^*}\left(1 - \exp\left(-\left(\frac{1}{\tau_c^*} + \frac{1}{\tau_e^*}\right)t\right)\right). \tag{7}$$

Here, the duty factor DF of a device under ac stress is defined as the probability of the transistor in accumulation mode that is effective for BTI stress (in some papers, DF is also referred to as the signal probability SP). The parameters τ_c^* and τ_e^* are defined as the effective capture and emission time constants under ac stress, which account for the duty factor effect

$$\frac{1}{\tau_c^*} = \frac{DF}{\tau_{c,\mathrm{Stress}}} + \frac{1 - DF}{\tau_{c,\mathrm{Relax}}} \tag{8}$$

$$\frac{1}{\tau_e^*} = \frac{DF}{\tau_{e,\mathrm{Stress}}} + \frac{1 - DF}{\tau_{e,\mathrm{Relax}}}. \tag{9}$$

Fig. 3 shows an example plot of the occupancy probability function, i.e., $P_c(DF, t, \vec{\tau})$, of a single defect as defined in (7), with the values of the time constants shown in the figure. The plot indicates that the occupancy probability P_c gradually increases with DF but rapidly rises with time at the range of 10^5 to 10^6 a.u.

Since the defect precursors (Si–Si bond in the SiO_2 dielectric according to [24]) are created in the fabrication process and uniformly distributed in the dielectric layer, the statistical distribution of capture/emission time constants associated with each defect is i.i.d. For each defect, the four components of $\vec{\tau}$ are correlated [4], and their joint distribution can be characterized for a specific technology. In this paper, we follow the assumptions in [17] to generate the distributions of time constants. Fig. 2 shows an example 2-D histogram of the joint distribution of $\tau_{c,\mathrm{Stress}}$ and $\tau_{e,\mathrm{Relax}}$, which are the dominant components of $\vec{\tau}$. The proposed approaches in this paper are independent of the distribution of $\vec{\tau}$.

We introduce the concept of the *mean defect occupancy probability*, i.e., $\bar{P}_c(DF, t)$, which captures the expected value of the probability of a defect charged with carriers (i.e., captured), based on the single-defect model of (7), and $f(\vec{\tau})$, the joint pdf of $\vec{\tau}$

$$\bar{P}_c(DF, t) = \int P_c(DF, t, \vec{\tau}) f(\vec{\tau}) d\vec{\tau}. \tag{10}$$

Fig. 4 shows an example of $\bar{P}_c(DF, t)$ function corresponding to the assumed $f(\vec{\tau})$ plotted in Fig. 2. This plot indicates that

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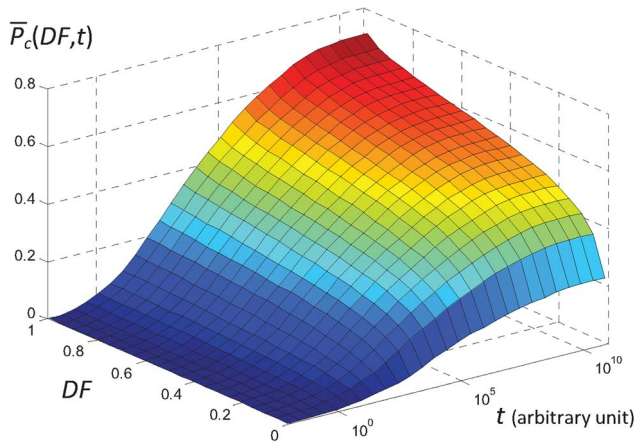

Fig. 4. Plot of mean defect occupancy probability function $\bar{P}_c(DF, t)$.

the mean occupancy probability is a monotonically increasing function of both DF and time. Due to averaging effects over large number of defects with different $\vec{\tau}$, $\bar{P}_c(DF, t)$ changes more gradually with time, as compared with P_c of a single defect in Fig. 3.

Since $\bar{P}_c(DF, t)$ is only determined by the distribution $f(\vec{\tau})$ and is independent of the circuit structure, it can be numerically precharacterized using (10) and stored in a look-up table for use in the circuit analysis.

For small-geometry devices, the number of defects in a MOS transistor is a relatively small number with relatively large variation [14]. For a transistor of length L and width W, the total number of oxide defects n is empirically modeled as a Poisson distribution [11], i.e.,

$$n \sim \text{Poisson}(N)$$
$$\text{where } N = N_{\text{ot}} W L. \tag{11}$$

Here, N_{ot} is the density of defects in the dielectric, and N is the total number of defects in the MOS transistor. Note that the Poisson distribution in (11) has similar form as the R–D model (4), but they are from different underlying mechanisms: R–D is modeled with interface traps (Si–H bond), while T–D is modeled with bulk oxide traps (missing oxygen atom in Si–O–Si bond [24]). Both kinds of traps are modeled as Poisson distributions due to the random location of the traps in small devices; however, these two distributions are not correlated in nature.

Similarly, the number of *occupied* defects, n_c, in a transistor also has a Poisson distribution,[1] with its mean value N_c calculated as follows:

$$n_c \sim \text{Poisson}\,(N_c)$$
$$\text{where } N_c = N \cdot \bar{P}_c(DF, t). \tag{12}$$

[1]The number of occupied defects in a device follows a Poisson distribution by definition: 1) each occupied defect has the same occurrence rate $N_c/(WL)$ within the device area of W by L; and 2) the occurrence of all occupied defects are independent with each other. This is similar to the number of all defects, which follows $n \sim \text{Poisson}(N)$ and is verified by experimental results in Section IV.

Observed in [15], the BTI-induced threshold degradation corresponding to each single defect follows an exponential distribution. Each defect $k = 1, \ldots, n$, contributes a threshold degradation of

$$\Delta V_{\text{th}}^{(k)} \sim \text{Exp}(\eta)$$
$$\text{where } \eta = \eta_0/(WL). \tag{13}$$

Like N_{ot}, η_0 is a technology-specific constant.

The total threshold voltage degradation ΔV_{th} of a transistor is the sum of contributions $\Delta V_{\text{th}}^{(k)}$ from all *occupied* defects k in the transistor, i.e.,

$$\Delta V_{\text{th}} = \sum_{k=1}^{n_c} \Delta V_{\text{th}}^{(k)}. \tag{14}$$

A closed form of this sum is derived in [15], and the PDF of ΔV_{th} turns out be to a complex distribution with mean $\mu = N_c\eta$ and variance $\sigma^2 = 2N_c\eta^2$. The mean value corresponds to the nominal case (i.e., each of N_c defects results in a threshold degradation of η). In [15], the probit plot of ΔV_{th} indicates that, for an adequate number of defects (e.g., $N_c \geq 10$), the transistor ΔV_{th} distribution can be approximated as a Gaussian by matching the mean and variance, resulting in the distribution

$$\Delta V_{\text{th-CT}} \sim N(N_c\eta, 2N_c\eta^2). \tag{15}$$

When the number of occupied defects N_c is sufficiently large, this Gaussian approximation is justified by central limit theorem (CLT), using the fact that the total threshold degradation is the sum of ΔV_{th} from each defect, which are i.i.d. exponential. For smaller devices with lower values of N_c, this Gaussian approximation is not necessarily accurate for individual devices, but the circuit-level timing analysis results still have good accuracy compared with MC simulation, which can be justified by the CLT because the circuit delay is the sum of the cell delays along the critical paths and approaches a Gaussian distribution. A more detailed discussion about this Gaussian approximation model is available in Section IV.

C. PVs *and Spatial Correlation*

Variations in the process parameters also contribute to BTI variability. PVs are typically classified as lot-to-lot, die-to-die (D2D), and within-die (WID) variations, according to their scope; they can be also categorized, based on their causes and predictability, as systematic or random variations. Some (but not all) WID variations exhibit spatial dependence knows as spatial correlation, which must be considered for accurate circuit analysis.

We employ a widely used variational paradigm, where a process parameter X is modeled as a random variable about its mean X_0 as [25], i.e.,

$$X = X_0 + \Delta X$$
$$\Delta X = X_g + X_s + X_r$$
$$\sigma_X^2 = \sigma_{X_g}^2 + \sigma_{X_s}^2 + \sigma_{X_r}^2. \tag{16}$$

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Here, X_g, X_s, and X_r stand for, respectively, the global component (from lot-to-lot or D2D variations), the spatially correlated component (from WID variation), and the residual random component of PVs. Under this model, all devices on the same die have the same global part X_g. The spatially correlated part is modeled using a widely used grid-based method [5] for the parameters that exhibit this property and is zero for those that are spatially uncorrelated. Under the spatial correlation model, the entire chip is divided into grids. All devices within the same grid have the same spatially correlated part X_s; the X_s parameters for devices in different grids are correlated, with the correlation falling off with the distance. The random part X_r is unique to each device in the system.

In this paper, we consider the variations in the transistor width W, the channel length L, the oxide thickness $T_{\rm ox}$, as well as shifts in the threshold voltage $V_{\rm th}$ due to RDFs. In other words, for each device, X represents elements of the set $\{W, L, T_{\rm ox}, V_{\rm th}\}$. As in the large body of work on SSTA, we assume Gaussian-distributed parameters for each of these process parameters, with W and L exhibiting spatial correlation and $T_{\rm ox}$ and $V_{\rm th}$ being uncorrelated from one device to the next. The spatial correlation structure is extracted as a correlation matrix [26], and processed using principal component analysis (PCA) to facilitate fast timing analysis [5]. The process parameter value in each grid is expressed as a linear combination of the independent principal components, with potentially reduced dimension.

Notationally, we express each process parameter X as a vector in a random space, with basis $\mathbf{e} = [\mathbf{e}_g, \mathbf{e}_s, \mathbf{e}_r, \epsilon]^{\mathbf{T}}$, as

$$\begin{aligned} X &= X_0 + \Delta X = X_0 + \mathbf{k}_X^{\mathbf{T}}\mathbf{e} \\ &= X_0 + \mathbf{k}_{Xg}^{\mathbf{T}}\mathbf{e}_g + \mathbf{k}_{Xss}^{\mathbf{T}} + \mathbf{k}_r^{\mathbf{T}}\mathbf{e}_r + k_\epsilon \epsilon \qquad (17) \\ \sigma_X^2 &= \mathbf{k}_X^{\mathbf{T}}\mathbf{k}_X, \mathrm{cov}(X_i, X_j) = \mathbf{k}_{Xi}^{\mathbf{T}}\mathbf{k}_{Xj} - k_{\epsilon_i}k_{\epsilon_j}. \qquad (18) \end{aligned}$$

Here, $\mathbf{e}_g = [e_{Wg}, e_{Lg}]^{\mathbf{T}}$ is the basis for the global part ($T_{\rm ox}$ variation and RDF effect are purely random, hence do not have a global part), $\mathbf{e}_s = [e_1, \ldots, e_t]^{\mathbf{T}}$ is the basis of principal components for the spatially correlated part, in which t is the number of dimensions after the PCA processing of correlated part, and $\mathbf{e}_r = [\epsilon_1, \ldots, \epsilon_m]^{\mathbf{T}}$ is the basis of random part. The dimension of random part m will depend on the implementation of the SSTA algorithm and can vary from constant to linear (of circuit size), as will be shown later in this paper. The random basis $\mathbf{e}_r$ and its coefficient vector $\mathbf{k}_r$ are implemented using a sparse data structure. The Gaussian variable $\epsilon \sim N(0, 1)$ is a separate independent random part for use in circuit-level timing analysis.

D. Consideration of PVs and BTI Interaction

PVs are created at manufacture time, while BTI degradation occurs during the circuit operation. Therefore the effect of PVs is independent from BTI, but the BTI effect will be impacted by PVs, i.e., the BTI degradation is dependent on the actual W, L, and $T_{\rm ox}$ of a transistor. This paper assumes the PVs and BTI effects (both R–D and CT models including variabilities) to be independent and uses a superposition model to calculate the total effect on circuit-level degradations. This is based on the following facts and considerations.

- The impact of PVs on BTI degradation is a second-order effect that is relatively small in nature.
- For W and L variations, [27] indicates the NBTI effect is more pronounced in narrow and long transistors. However, the transistors on the critical paths are normally sized larger (wider) for timing performance, hence less affected by the W and L variations.
- For $T_{\rm ox}$ variation, a smaller $T_{\rm ox}$ causes elevated BTI degradation speed, but also gives smaller initial $V_{\rm th}$ and propagation delay. Therefore, the interaction effect actually cancels out with each other to some degree and ignoring it yields pessimistic and safe approximations.
- The independent assumption simplifies the modeling and analysis and helps achieve linear computational complexity and good scalability (see Section III-C).

III. Timing Analysis Under Variations

This section introduces the logic cell delay model under BTI variations and PVs. Based on this model, a scalable approach for statistical timing analysis of large digital logic circuits is outlined.

A. Cell Timing Model and Characterization

We use a cell delay degradation model that is similar to [28]. The delay d_i of cell i is modeled using a first-order Taylor approximation as a linear function of process parameters W_j, L_j, and $T_{\text{ox-}j}$ of each transistor j in cell i, and BTI degradation $\Delta V_{\rm th}^{\rm jj}$ of each transistor j:

$$d_i = d_{i0} + \Delta d_i = d_{i0} + \sum_{X \in \mathbf{P}_i} \frac{\partial d_i}{\partial X} \Delta X.$$

Here, $\mathbf{P}_i = \{W_j, L_j, T_{\text{ox-}j}, V_{\rm th}^{\rm jj}\}$, $j \in$ cell i is the set of variational parameters. The nominal propagation delay d_{i0} and its sensitivity $\partial d_i / \partial X$ to parameter $X \in \mathbf{P}_i$ are computed using standard techniques through SPICE simulations. This part of the calculation is circuit independent and performed as part of library characterization.

Since all variational parameters $X \in \mathbf{P}_i$ are expressed as vectors in the random variable space $\mathbf{e}$ in Section II-C, d_i, which is a linear combination of these parameters, is also a vector in space $\mathbf{e}$

$$\begin{aligned} d_i &= d_{i0} + \left(\sum_{X \in \mathbf{P}_i} \frac{\partial d_i}{\partial X} \mathbf{k}_X \right)^{\mathbf{T}} \mathbf{e} \\ &= d_{i0} + \mathbf{k}_{d_g}^{\mathbf{T}}\mathbf{e}_g + \mathbf{k}_{d_s}^{\mathbf{T}}\mathbf{e}_s + \mathbf{k}_{d_r}^{\mathbf{T}}\mathbf{e}_r. \qquad (19) \end{aligned}$$

Here, the random part $\mathbf{e}_r = \{\epsilon_X\}_{X \in \mathbf{P}_i}$ is extended to include the random parts from all variational parameters $X \in \mathbf{P}_i$ in cell i.

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TABLE I
T-SSTA RESULTS UNDER VARIATIONS (TIME UNIT: PS)

c3540 16nm	MC		T-SSTA1		T-SSTA2		T-SSTA3	
D_0=582.3	$\mu_{\Delta D}$	$\sigma_{\Delta D}$	$\mu_{\Delta D}$	$\sigma_{\Delta D}$	$\mu_{\Delta D}$	$\sigma_{\Delta D}$	$\mu_{\Delta D}$	$\sigma_{\Delta D}$
BTI RD	23.8	3.6	23.8	3.7	23.8	3.7	23.8	3.7
BTI CT	29.8	8.9	29.7	8.8	29.6	8.8	29.6	8.8
PV	0.5	14.8	4.2	14.1	0.3	14.6	1.3	14.6
ALL	53.9	17.2	57.1	17.0	54.0	17.5	54.8	17.4

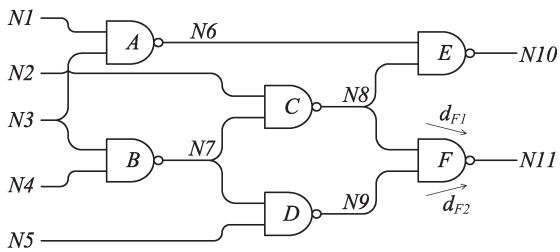

Fig. 5. Example circuit showing path reconvergence.

B. Circuit-Level Timing Analysis

At the circuit level, timing analysis is performed using a PERT-like traversal [5] at a fixed time point, where the contributions of the temporal BTI variations can be characterized using the models described in Sections II-A and II-B. The $V_{\rm th}$ degradation due to these two models are uncorrelated and are found to substantially affect the circuit-level delay.

In our initial implementation of algorithm, as in [7], the random part $\mathbf{k}_r^T\mathbf{e}_r$ of arrival time is merged into the separate independent term $k_\epsilon\epsilon$ that is the product of scalars to reduce the computational complexity. The T-SSTA result of this method is denoted as T-SSTA1. Table I shows the mean and standard deviation (SD) of circuit delay degradation on benchmark c3540 under a 16-nm technology model at $t = 2000$ (a.u.), splitting the contribution of the mean and variance into those attributable to BTI R–D, BTI CT, PVs, and finally, presenting the combined values (ALL). The results of mean and SD calculated by MC simulation are listed as reference, and the results indicate that the mean value of delay degradation is mainly contributed by BTI RD and BTI CT, while the SD is mainly contributed by BTI CT and PV effects. By comparing results for "ALL" from T-SSTA1 method with MC, we can see T-SSTA1 overestimates the mean value and underestimates the SD, where the errors are mainly coming from the BTI CT part.

Investigating this further, we find that the error between conventional method (T-SSTA1) and MC simulation can be attributed to the correlations that arise due to path reconvergence, which are neglected in T-SSTA1. The BTI CT part of $V_{\rm th}$ degradation contains a significant amount of independent random component in form (17), hence generates large errors due to path reconvergence. We illustrate the path reconvergence effect through Fig. 5, which shows an example circuit where the arrival time AT of node $N11$, denoted as AT_{N11}, is calculated as follows:

$$AT_{N11} = \max(AT_{N8} + d_{F1}, AT_{N9} + d_{F2}) \tag{20}$$

where AT_{N8} and AT_{N9} stand for the arrival times of node $N8$ and $N9$, while d_{F1} and d_{F2} stand for the delays from the first and second input to the output of cell F. The arrival times and cell delays are modeled as vectors in random variable space $\mathbf{e}$. Note that since cell C and D have a common fan-in of cell B, AT_{N8} and AT_{N9} are both dependent on the random component of the parameters of cell B, corresponding to the impact of X_r in (16). As a result, the independent components in the expression for AT_{N8} and AT_{N9} are not independent of each other but are correlated. However, the conventional SSTA method, using an separate independent term $k_\epsilon\epsilon$ to replace $\mathbf{k}_r\mathbf{e}_r$, does not capture this path correlation and introduces errors. The same situation occurs when calculating the total delay using the maximum of AT_{N10} and AT_{N11}, which are correlated because the paths from node $N8$ reconverge.

One natural way to resolve this problem is to preserve the entire random part $\mathbf{k}_r\mathbf{e}_r$ when calculating the arrival times, by which the path correlation is captured. This method is denoted as T-SSTA2 in Table I, and the results indicate this method is much more accurate than T-SSTA1. However, the cost paid for this accuracy is in the increased computation time associated with the growing size of the random part (e.g., AT_{N11} in the example contains components from cells B, C, D, and F). The computational complexity is discussed with more details in Section III-C, and the experimental runtime and storage comparison will be given in Section IV.

We employ a third method, denoted as T-SSTA3 in Table I, taken from [29], to provide a tradeoff between the accuracy and complexity. This removes only the smaller elements in the random vector $\mathbf{k}_r$ using preset threshold and merges them into the separate term k_ϵ. Results in Table I show that this method achieves good accuracy (within 2% error compared with T-SSTA2 and Monte Carlo) with low computation. We will expand on this in Section IV.

C. Computational Complexity

To calculate the circuit delay, the SSTA algorithm does a topological traversal through the digital circuit. For each node (logic cell), the timing analysis performs k sum-of-two and $k-1$ max-of-two operations, where k is the number of fan-in of the cell. In random space $\mathbf{e}$ with dimension d, the numbers of total sum and max operations for SSTA are

$$N_{\rm sum} = n \cdot k \cdot d \tag{21}$$

$$N_{\rm max} = n \cdot (k-1) \cdot d. \tag{22}$$

Here, $d = d_g + d_s + d_r$, in which d_g, d_s, and d_r are the dimensions of global component $\mathbf{e_g}$, spatial component $\mathbf{e_s}$, and random component $\mathbf{e_r}$, respectively. The values of d_g and d_s are well bounded by PCA algorithm and, therefore, can be regarded as constant. The values of d_r depend on how the random part is handled, as discussed in Section III-B. For methods T-SSTA1 and T-SSTA3, d_r is bounded by a constant, while for T-SSTA2, d_r can significantly grow depending on the circuit size and structure. For simplicity, it can be roughly approximated as $d_r \propto \sqrt{n}$, which corresponds to the depth of the circuit (number of cells on the critical paths). Therefore, the computational complexity is $O(n)$ for T-SSTA1 and T-SSTA3 and $O(n^{1.5})$ for T-SSTA2. This result indicates the proposed T-SSTA3 method has good scalability to handle large-scale circuits.

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TABLE II
MEAN AND SD OF CIRCUIT DELAY USING DIFFERENT METHODS (TIME UNIT: PS, AVERAGE ERROR SHOWN FOR $\mu_{\Delta D}$ AND σ_D)

Circuit & Technology		Initial	T-SSTA1		T-SSTA2		T-SSTA3		MC		$V_{\text{th-RD}}$(mV)		$V_{\text{th-CT}}$(mV)	
		D_0	μ_D	σ_D	μ_D	σ_D	μ_D	σ_D	μ_D	σ_D	μ	σ	μ	σ
32 nm	c2670	734	802	13.3	795	14.6	796	14.4	794	14.7	17.8	3.3	16.9	6.1
	c3540	812	910	15.6	903	16.7	904	16.7	903	16.5	17.7	3.1	16.6	5.7
	c5315	666	736	12.9	735	13.1	735	13.1	735	12.9	18.1	3.2	17.5	6.0
	c6288	1416	1580	23.5	1574	24.1	1576	24.0	1574	23.6	17.3	2.9	16.5	5.3
	c7552	650	714	11.6	709	12.5	710	12.5	710	12.4	18.2	3.3	17.6	6.2
	b15	1416	1580	24.2	1571	26.2	1574	25.7	1572	25.9	17.0	3.1	16.5	5.6
	b17	1634	1770	27.0	1750	29.1	1757	28.5	1752	28.7	16.6	3.1	16.1	5.5
	b20	1432	1566	24.1	1554	26.8	1555	26.7	1555	26.3	17.8	3.2	17.3	6.0
	b21	1598	1765	27.0	1757	29.9	1758	29.7	1758	30.5	17.8	3.3	17.4	6.1
	b22	1520	1655	23.4	1645	25.3	1646	25.1	1645	25.3	17.4	3.2	17.0	6.0
Avg Err %			7.24	6.19	0.46	1.36	1.22	1.36						
22 nm	c2670	617	669	12.6	663	13.8	664	13.6	663	13.9	18.1	4.6	17.4	9.0
	c3540	671	760	15.1	754	16.5	755	16.4	754	16.7	17.6	4.3	16.7	8.1
	c5315	557	625	11.9	624	12.2	624	12.1	624	12.3	17.7	4.4	17.0	8.4
	c6288	1211	1366	22.8	1359	23.5	1362	23.3	1359	23.6	17.3	4.0	16.6	7.7
	c7552	549	607	10.6	601	12.3	602	12.2	601	12.3	17.9	4.5	17.3	8.8
	b15	1151	1287	23.9	1274	27.4	1275	27.0	1275	26.4	16.8	4.3	16.2	8.0
	b17	1312	1444	23.7	1437	24.4	1443	23.4	1440	25.4	16.5	4.3	16.0	7.9
	b20	1144	1293	23.4	1273	27.8	1274	27.6	1274	28.0	18.1	4.5	17.6	8.9
	b21	1251	1392	25.0	1388	26.9	1388	26.8	1388	26.6	18.0	4.5	17.6	8.9
	b22	1252	1379	23.0	1371	24.7	1372	24.6	1372	24.9	17.7	4.5	17.4	8.8
Avg Err %			7.48	8.45	0.75	1.50	1.30	2.16						
16 nm	c2670	537	592	13.9	582	15.7	584	15.4	582	15.9	18.5	6.2	17.9	12.7
	c3540	582	657	17.5	652	18.1	653	18.0	652	18.9	17.0	5.4	16.1	10.7
	c5315	489	570	14.2	567	14.6	568	14.5	568	15.1	18.2	5.8	17.7	12.0
	c6288	1092	1253	26.2	1247	26.7	1251	26.5	1247	26.6	17.3	5.3	16.6	10.5
	c7552	480	559	12.9	545	16.8	548	16.6	546	16.9	18.2	6.0	17.6	12.4
	b15	986	1143	26.0	1121	31.9	1124	31.2	1125	33.1	16.8	5.6	16.3	11.0
	b17	1100	1318	28.6	1293	33.2	1299	32.8	1293	33.6	16.7	5.6	16.2	11.0
	b20	941	1083	23.6	1075	23.5	1080	23.0	1078	25.2	17.9	5.9	17.4	12.1
	b21	1038	1158	26.7	1148	30.5	1149	29.8	1149	31.1	17.2	5.8	16.7	11.6
	b22	1072	1219	25.8	1206	29.0	1207	28.7	1208	29.1	17.9	6.0	17.5	12.2
Avg Err %			9.97	11.95	0.97	2.39	1.53	3.64						
Total Avg Err			8.23	8.86	0.73	1.75	1.35	2.39						

IV. RESULTS

Our approach for timing analysis under BTI variations and PVs is applied to ISCAS85 and ITC99 benchmarks. The benchmark circuits are mapped to a subset of the Nangate cell library [30] using ABC [31], with placement carried out using a simulated annealing algorithm. The benchmark circuits are scaled down to 32, 22, and 16 nm for comparisons under different technology models. The characterization of cell delay and of its sensitivities to variational parameters is performed using HSPICE simulation under PTM models [22]. Both the proposed analytical method and the MC method (for verification) are implemented in C++ and run on a Linux PC (3GHz CPU, 2GB RAM).

The PVs in W, L, and T_{ox} are set to $3\sigma = 4\%$ of their mean values [32]. The V_{th} variation due to RDF is dependent on the device size [20]. It has a Gaussian distribution with mean value $\mu = 0$, and standard variation

$$\sigma_{V_{\text{th}}} = \sigma_{V_{\text{th0}}} \sqrt{\frac{W_0 L_0}{WL}} \tag{23}$$

in which $\sigma_{V_{\text{th0}}}$ is the RDF-induced threshold SD of a minimum-sized device (W_0 by L_0). The value of $\sigma_{V_{\text{th0}}}$ is dependent on process parameters T_{ox} and N_a, as well as the doping profile of the channel [20]. Here, we assume $3\sigma_{V_{\text{th0}}} = 5\%$ of the nominal V_{th}. The parameter variations of W and L are split into 20% of global variation, 20% of spatially correlated variation, and 60% of random variation, while the variations of T_{ox} and V_{th} are fully random. The grid-based spatial correlation matrix is generated using the distance-based method in [26], with the number of grids growing with circuit size, as shown in Table III.

The MC simulation framework for verification of the proposed approach is set up as follows: the simulation program randomly generates 5000 circuit instances (we found it a good tradeoff of accuracy and runtime). For each circuit instance, the ΔV_{th} of each MOS transistor is calculated as the sum of the following three components:

(a) $\Delta V_{\text{th-RD}}$, which is set by (5) and is randomly generated based on the distribution of ΔN_{IT} as specified in (4);
(b) $\Delta V_{\text{th-CT}}$, which is set as the sum of ΔV_{th} of all defects that are randomly generated using distributions (11) and (13);
(c) $\Delta V_{\text{th-RDF}}$, which is due to RDF effects and set by (23).

The contributions of $\Delta V_{\text{th-RD}}$ and $\Delta V_{\text{th-CT}}$ vary with different technologies [3]. In the experiments, it is assumed that these two have comparable mean values, so that their contributions to circuit-level variations can be easily visualized. The process parameters W, L, and T_{ox} of each MOS transistor are also generated according to their distributions and correlation models. Then, the propagation delay of each cell is calculated using (19) and precharacterized cell delay and sensitivity data. Based on these values and a PERT-like traversal, the total delay of the circuit instances is evaluated using SSTA.

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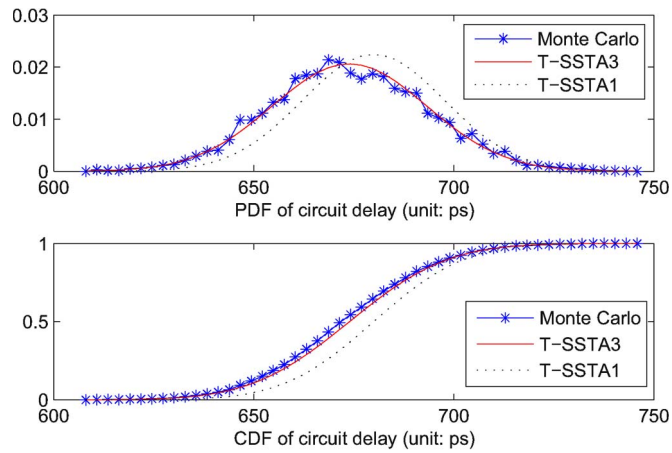

Fig. 6. Delay PDF and CDF of c3540 with 16-nm model.

For each benchmark circuit, the mean and SD of the circuit delay are calculated at time $t = 2000$ (a.u.), using the proposed analytical method and MC simulation. The three methods of handling random parts discussed in Section III-B are separately implemented. As before

- T-SSTA1 merges the random part into one variable;
- T-SSTA2 preserves full random part;
- T-SSTA3 partially lumps the random part.

Table II shows the nominal delay D_0 of each benchmark circuit, as well as the mean μ and SD σ of the circuit delay using three analytical methods and the MC simulation at 32, 22, and 16 nm. The last row shows the relative error of $\mu_{\Delta D}$ and σ_D of each analytical method, as compared with MC.

The mean and SD of the $\Delta V_{\rm th}$ contributions (averaged over all devices in the circuit) from the R–D model and from the CT model are also listed in Table II. Note that the simulation is based on the assumption that the $\Delta V_{\rm th}$ contributions (mean value) from R–D model and CT model are comparable. This assumption is made to give a general insight that the CT model predicts significantly larger BTI variability than R–D model. The proposed approaches for circuit degradation analysis is actually independent with this assumption and can handle different cases of the BTI degradation model. In general cases of application, both R–D and CT models of BTI effects can be characterized for given technology and used for analyzing the circuit timing degradations.

It is also worth noting that under certain cases (particularly at 16 nm, under the CT model), the value of 3σ can be larger than μ, indicating Gaussian distribution may not be an accurate approximation of $\Delta V_{\rm th}$ since $\Delta V_{\rm th}$ from BTI effects should always be positive. However, this inaccuracy of the Gaussian approximation is averaged out by the sum of delay along the critical path, and the circuit-level delay, calculated by sum and max operations in SSTA, and approaches a Gaussian distribution according to the CLT, which does not require the transistor $\Delta V_{\rm th}$ to be Gaussian. Therefore, the proposed method appears to be robust even under this model inaccuracy, as verified by the good accuracy indicated in Table II, and the visually verified match between distribution functions plotted in Fig. 6, which shows an example of the circuit delay distribution for c3540 at 16 nm at $t = 2000$ (a.u.). The T-SSTA3 and MC methods match well, verifying the validity of our assumptions; T-SSTA1 is significantly different due to the omission of path correlations.

TABLE III
COMPARISON OF COMPUTATIONAL COMPLEXITIES, WHERE $T_{\rm exe}$ = RUNTIME, [CELLS] = AVERAGE NUMBER OF CORRELATED CELLS

Circuit	Size		T-SSTA1	T-SSTA2		T-SSTA3		MC
Name	#cells	#grids	T_{exe}	T_{exe}	[Cells]	T_{exe}	[Cells]	T_{exe}
c2670	759	16	3.4s	5.8s	26.2	6.7s	3.0	108s
c3540	1033	16	5.7s	13.5s	109.0	12.8s	2.7	201s
c5315	1699	16	7.2s	14.1s	40.8	15.2s	2.9	261s
c6288	3560	64	17.1s	137.8s	473.6	38.9s	2.8	627s
c7552	2361	36	9.8s	21.1s	53.7	20.3s	3.2	352s
b15	6548	100	34.8s	352.4s	512.1	89.6s	3.0	1181s
b17	20407	361	109.3s	1513s	421.6	306.0s	3.5	3645s
b20	11033	169	55.2s	482.1s	362.3	139.0s	3.4	1926s
b21	10873	169	52.9s	439.1s	351.4	133.5s	2.8	1845s
b22	14794	225	72.4s	671.2s	304.9	188.1s	3.1	2507s

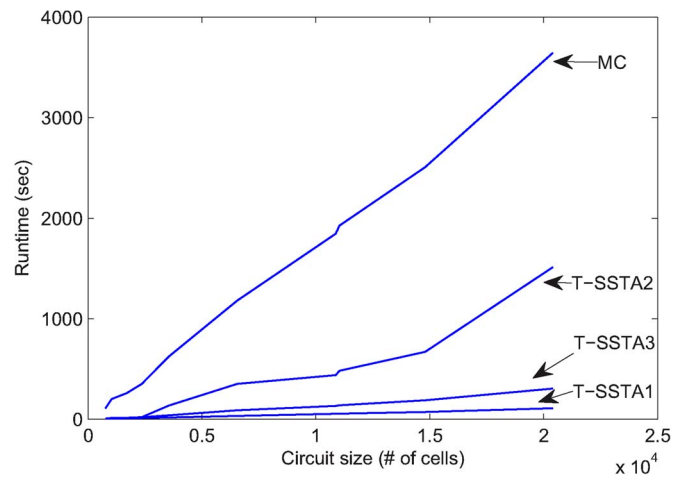

Fig. 7. Runtime versus circuit size of different methods.

Table III compares the runtime and storage complexity (in terms of the average number of correlated cells, denoted as [cells]) of the analytical methods and MC. Fig. 7 shows the runtime versus circuit size (number of logic cells) for the different methods.

The results indicate that the runtime of partially lumping random part (T-SSTA3) method linearly grows with circuit size increasing, indicating good scalability. It has an overall error of about 2% to MC and is 15× faster on average. Furthermore, it reduces runtime by 60% and storage by 98% on average compared with T-SSTA2, with similar accuracy. The conventional method (T-SSTA1) has the shortest runtime but has nearly 9% errors with respect to MC. The results also verify that the Gaussian approximations for $\Delta V_{\rm th}$ in BTI R–D and CT models are valid; the method is accurate, efficient, and scalable. Moreover, the SD of circuit delay σ_D increases with technology downscaling, indicating that random timing variation attributable to BTI is a growing issue.

Fig. 8 shows the variance of circuit delay that originates from PVs, R–D BTI variations, and CT BTI variations separately, for different benchmarks under the 32-, 22-, and 16-nm technology models. For better presentation of data, the variances are normalized to the total variance of 32-nm model for each benchmark. These results indicate that the CT model

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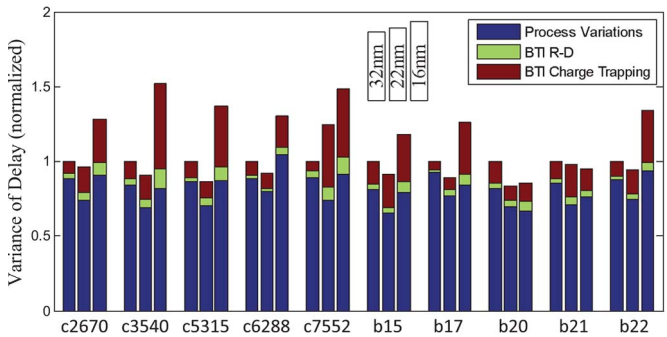

Fig. 8. Relative contribution of BTI CT, BTI R–D, and PV to circuit delay variation.

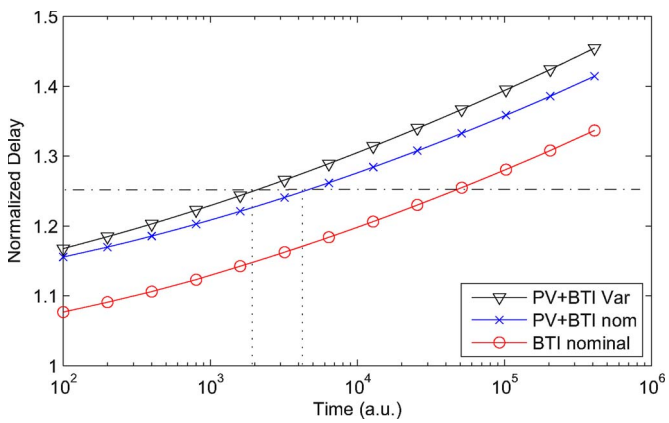

Fig. 9. Delay degradation versus time of c5315.

is the dominant component of BTI variations and makes a significant contribution to circuit delay variation. In contrast, the BTI variations under the R–D model only introduce a relatively small portion of delay variations. Unlike PVs, which have nearly constant influence on delay variation, the impact of BTI variations grows with scaling, thus becoming increasingly severe in future.

Furthermore, according to the results in Table II and Fig. 8, the circuit-level delay variation that can be attributed to BTI variations is not as significant as the single-device $\Delta V_{\rm th}$ variation due to BTI effect of a small transistor shown in Fig. 1(b). This is mainly due to the facts that (a) transistors on the critical paths usually have larger than minimum sizes to help with timing; and (b) the average out of randomness of the transistor $\Delta V_{\rm th}$ on the critical paths due the sum of delay.

Fig. 9 presents the circuit delay degradation versus time curves of benchmark c5315 at 16 nm. Three curves are shown for the normalized delay of (1) nominal BTI degradation, without any variation model; (2) $\mu + 3\sigma$ of PV and nominal BTI degradation; and (3) $\mu + 3\sigma$ of PV and BTI with variabilities (under both R–D and CT models). The results indicate that BTI degradation and variability, which grow with time, make up the dominant part of total delay degradation, particularly at the later point of circuit lifetime. Furthermore, BTI variations has a significant impact on circuit reliability. In this case, the circuit lifetime will be overestimated by over $2\times$ if BTI variations is not considered (lifetime defined as 25% increase in delay from time zero).

V. Conclusion

This paper incorporates both the R–D and CT models of BTI variations into a T-SSTA framework, capturing PVs and path correlations. Experimental results show that the proposed analysis method is fast and accurate. Our results indicate that the CT mechanism, which has been neglected by the EDA field so far, is the dominant source of BTI variations with significant and growing contributions to circuit timing variations.

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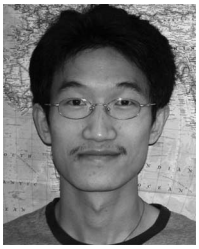

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